



Dynamic stabilization and feedback control of the pendulum in any desired position

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ABSTRACT

One example of stabilizing a statically unstable system is the dynamic stabilization of the pendulum by means of oscillations of its suspension point. The pendulum whose suspension point oscillates in the vertical direction which stabilizes it in the inverted position is the so-called Kapitza pendulum. It turns out that it is possible to stabilize the pendulum in other positions by choosing the right direction of oscillations of the pendulum's suspension point. In addition, if the direction of oscillations of the pendulum suspension point can change over time, some interesting control problems may arise, such as transition of the pendulum from point to point or active damping control. This paper presents the experimental results that confirm the occurrence of the above-mentioned processes in practice. The work also describes the analytical approximation of the pendulum motion. Moreover, it turns out that the results obtained with this approximation largely agree in the experiment, which was also presented in this work.

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1. Introduction

One of the methods of stabilizing the statically unstable systems are the so-called parametric excitations. Over a century ago, Andrew Stephenson [1,2] noticed that the fast vertical oscillations of the pendulum's suspension point stabilized the pendulum in an inverted position, so the originally unstable equilibrium position became a stable position. In 1932, Lowenstern [3] derived a general equation of motion for an inverted pendulum with an oscillating pivot but still did not explain why the pendulum is stable. In a groundbreaking work [4], Kapitza gave the explanation why such a system is stable - fast oscillations of the pendulum's suspension point lead to form the effective potential, whose minimum appears for the pendulum's vertical position. This work initiated a series of studies on the stabilization of statically unstable systems using the parametric excitations. The Kapitza method can be used for example to increase the stability of elastic systems by means of vibrations [5–7]. Zelei and Stepan [8] found an interesting application of the dynamic stabilization method: they used vertical periodic force to stabilize a small ship. It turns out that the highly dynamic gliding behavior of flying snakes, which is dominated by an undulation can be also explained by Kapitza method [9].

The dynamic stabilization method works equally well at both a micro and nano scale. Richards et al. [10] showed that the colloidal particle suspended in water and trapped by modulated with high frequency optical tweezers behaves like a Kapitza pendulum, so dynamic stabilization could be exploited in applications aimed at particle separation at small scales. Another interesting application of the Kapitza method is dynamic stabilization mechanism for plasma instabilities by strong

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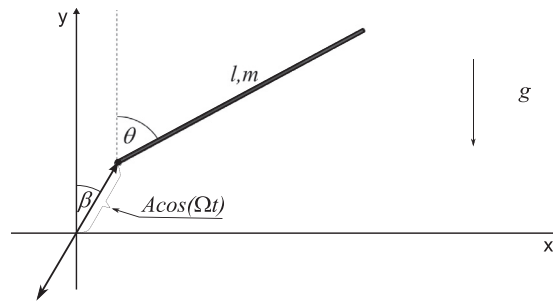


Fig. 1. Physical model of pendulum.

driving force oscillation [11]. Dynamic stabilization can be also found in other atomic and quantum processes [12,13], as well as in control theory in many application areas (see for example, [14–16]).

The vertical oscillations of the pendulum suspension point cause stabilization in the inverted position, while the horizontal oscillations stabilize the pendulum between the vertical down and the horizontal position [17]. It turns out that it is possible to generalize the problem to dynamic stabilization of the pendulum in an arbitrary position. This can be achieved by means of oscillations of the pendulum's suspension point at the appropriate angle (not only vertical or horizontal oscillations). In [18] Sudor and Bishop numerically examined the case when the direction of oscillations of the pendulum's suspension point was slightly tilted from the gravity direction. In [19] Weibel et al. theoretically studied the effects of rapid periodic forcing on the dynamics of the pendulum on the cart in the case when the tilt of direction of oscillations of the pendulum's suspension point was large. They showed the existence of a periodic orbit near each equilibrium position. In [20] swing-up and the stabilization at the upright position of the free link in a two-link underactuated manipulator was theoretically and experimentally tested. A two-link manipulator was moving in the plane slightly inclined from the horizontal plane. The first link was directly actuated, and the second link was freely connected to the end of the first link. The oscillations the first link around desired position produced oscillations of the suspension point of the second link (pendulum) - suspension point moved along a fragment of the arch. The swing-up of the second link was realized by changing the desired position of the first link (from 0 to $\pi/2$) under its constant excitation. Finally, the reachable and stabilizable area of a two-link underactuated manipulator was clarified theoretically as well as experimentally by Yabuno et al in [21]. A The two-link manipulator previously described, in this case was moving in the vertical plane. They numerically determined conditions for stable equilibrium positions of the free link and proposed a strategy of the motion control to the desired position of the free link: using such a control they showed the possibility to stabilize the pendulum in any position and to transfer the pendulum from point to point. The pendulum where the direction of oscillations of its suspension point was tilted and the suspension point oscillated in a straight line was numerical tested in [22]. In [23] the analytically obtained conditions of the pendulum stability were presented. These conditions concern relationships between physical parameters of the system, angle of oscillation of the pendulums suspension point and desired pendulum position. Mechanical system where the pendulum (perfectly rigid rod with mass m and length l) is driven by periodic oscillations of the suspension point and the direction of these oscillations is equal β is shown in Fig. 1.

The pendulum equilibrium position depends on the direction of oscillation of the suspension point determined by angle β and it turns out that any position of the pendulum in the range $(0, 2\pi)$ can be a stable position. Should this be the case, the ability to change the β during stabilization gives the opportunity to transition the pendulum from point to point. This process of control can be called the dynamic stabilization of the pendulum in a moving potential well because the change in the oscillation angle of the pendulums suspension point changes the effective potential of the pendulum and thus the location of its minimum. The numerical results indeed confirm this possibility [24].

A dynamically stabilized pendulum deflected from its equilibrium position starts to oscillate around this position therefore the pendulum is Lyapunov stable but not asymptotically stable. To suppress these excess vibrations, the angular velocity of β as the control input can be used, as confirmed numerically in the work [25] where the full state feedback control was used.

The previous studies on dynamic stabilization of the pendulum in any desired position have mainly focused on the problem of pendulum stability however, the mathematical description of the pendulum's motion and its oscillations has not yet been studied carefully. The case when the pendulum suspension point oscillates in an arc has already been experimentally verified in [21], whereas the case when the pendulum suspension point oscillates in a straight line has not been verified.

The main purposes of this work therefore are:

- to find and experimentally verify the time and frequency domain description of the pendulum motion during dynamic stabilization.
- to experimentally verify the case when the pendulum suspension point oscillates in a straight line.

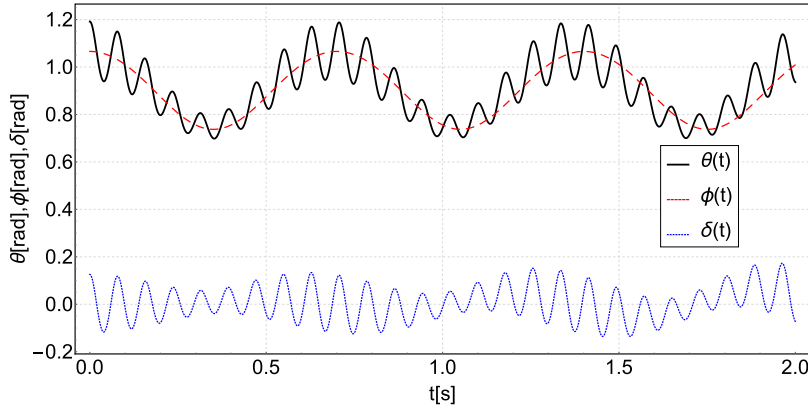


Fig. 2. Simulation results of pendulum's motion.

For the purposes of the experiment, the pendulum system that allows to change the direction of oscillations of the pendulum suspension point was made. The built system measures the position of the pendulum and the direction of oscillations (angle β).

This paper is organized as follows. Section 2 presents the mathematical model of the pendulum and briefly describe stability analysis. The approximation solution of the pendulum motion during dynamic stabilization is described in Section 3. Mathematical description of the pendulum motion in time and frequency domain was derived for the case of an undamped as well as damped pendulum. Section 4 contains all details concerning the experimental setup: description of the mechanical parts and hardware used during the experimental verification of dynamical stabilization of the pendulum. The results obtained during the experiments compared with the simulation and approximation results are presented in Section 5. Finally, the last section concludes this paper.

2. Mathematical model and its stability

Using for example the Lagrange formalism, the equation of motion of the pendulum presented in Fig. 1 can be formed as:

$$\ddot{\theta} = \frac{3(A\Omega^2 \sin(\beta - \theta) \cos(t\Omega) + g \sin(\theta))}{2l} \quad (1)$$

where l is the pendulum length, A is the amplitude of oscillations of the suspension point, Ω - the frequency of these oscillations, β - angle determining the direction of oscillations of the suspension point. Fig. 2 shows the numerical solution of the Eq. (1) for values: $l = 0.19$ m, $A = 0.03$ m, $\Omega = 80$ rad/s, $g = 9.81$ m/s², $\beta = \pi/6$ rad, $\theta(0) = 1.20$ rad, $\dot{\theta}(0) = 0$ rad/s. Due to the fact that the pendulum moves in a constant gravitational field and additionally a fast-changing force causes vibrations of its suspension point, the pendulum will move under the influence of gravity and additionally execute small oscillations as can be seen in Fig. 2.

The motion of the pendulum therefore can be separated into two parts:

$$\theta(t) = \phi(t) + \delta(t) \quad (2)$$

where $\phi(t)$ describes the "smooth" movement of the pendulum, averaged due to the rapid oscillations. At this point, the Kapitza approach can be applied, i.e. find the equation describing the smooth motion of the pendulum. Substituting Eq. (2) into (1) and expanding the result in the first-order Taylor series with respect to small oscillations $\delta(t)$, the equation describing the "smooth" motion (averaged due to fast oscillations) of the pendulum can be presented as following:

$$\ddot{\phi} = \frac{9A^2\Omega^2 \sin(2(\beta - \phi))}{16l^2} + \frac{3g \sin(\phi)}{2l} \quad (3)$$

The detailed derivation of the Eq. (3) can be found in [23] and also in Section Appendix A.

Eq. (1) is nonlinear and furthermore depends on time explicitly, therefore the stability analysis is based on the Eq. (3) describing "smooth" motion.

When forces acting on a mechanical system are conservative, system is in stable equilibrium state if the value of its total potential energy is a minimum (see, e.g [26]). The force driving the pendulums suspension point as well as the gravity force is a conservative force, therefore the total force acting on the pendulum can be written as the negative gradient of a potential energy:

$$F = -\nabla U_{ef} \quad (4)$$

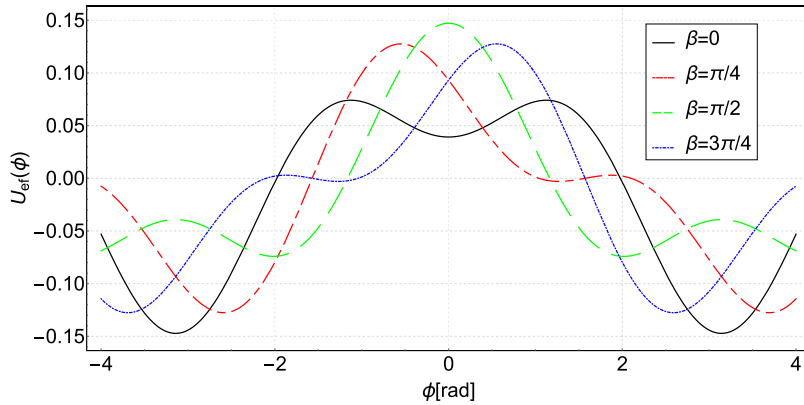


Fig. 3. Effective potentials of the pendulum.

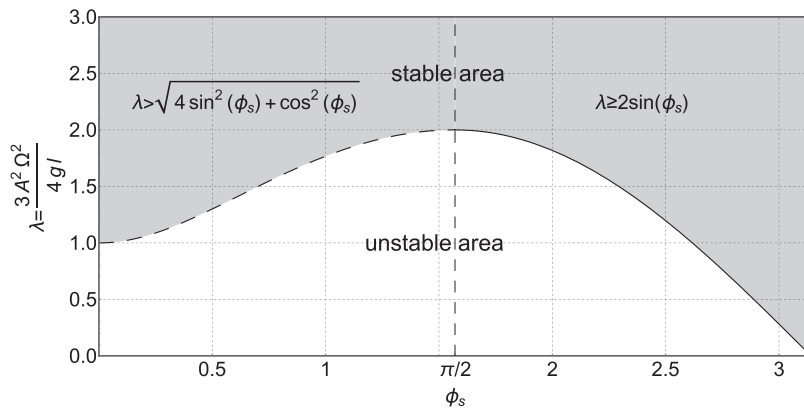


Fig. 4. Stability diagram of the pendulum.

where U_{ef} is the effective potential energy. In the case of physical pendulum, according to Newton's second law, the Eq. (4) can be written as:

$$\frac{1}{3}ml^2\ddot{\phi} = -\frac{dU_{ef}}{d\phi} \Rightarrow U_{ef} = -\frac{1}{3}ml^2 \int \ddot{\phi} d\phi \text{ which allows to determine the effective potential function:}$$

$$U_{ef} = \frac{1}{2}glm \cos(\phi) - \frac{3}{32}A^2m\Omega^2 \cos(2(\beta - \phi)) \quad (5)$$

Fig. 3 presents the effective potential (Eq. (5)) for different directions of oscillations of the suspension point. It shows that a change in β causes a change in the position of the potential minimum, and thus a change in the pendulum's stable position. The pendulum position $\phi = \phi_s$ is a stable position if the effective potential has a local minimum for $\phi = \phi_s$. The first derivative of the effective potential equated to zero gives the first condition for a local minimum $\beta(\phi_s)$. The second derivative of the potential greater than zero gives the final condition for a local minimum of the effective potential. The stability conditions of the pendulum in position $\phi = \phi_s$ can therefore be written in the form:

$$\begin{cases} \beta = \beta_s = \frac{1}{2}(2\phi_s - \arcsin(\frac{2\sin(\phi_s)}{\lambda})) \\ \lambda > \sqrt{4\sin^2(\phi_s) + \cos^2(\phi_s)} & \text{for } \phi_s \in (0, \frac{\pi}{2}) \\ \lambda \geq 2\sin(\phi_s) & \text{for } \phi_s \in (\frac{\pi}{2}, \pi) \end{cases} \quad (6)$$

where $\lambda = \frac{3A^2\Omega^2}{4gl}$.

Based on the Eq. (6), the pendulum stability diagram can be drawn. Fig. 4 shows the stability and instability area of the pendulum depending on the selected equilibrium point ϕ_s and parameter λ which describes the physical properties of the system.

As can be seen in Fig. 4 to achieve the stability in any position ϕ_s , the parameter λ must be greater than 2.

Considering the parameter values used in the example shown in Fig. 2, the stability parameter $\lambda = 2.32$ therefore meets the stability conditions for the whole range of ϕ_s . The detailed analysis of the pendulum stability can be found in [23].

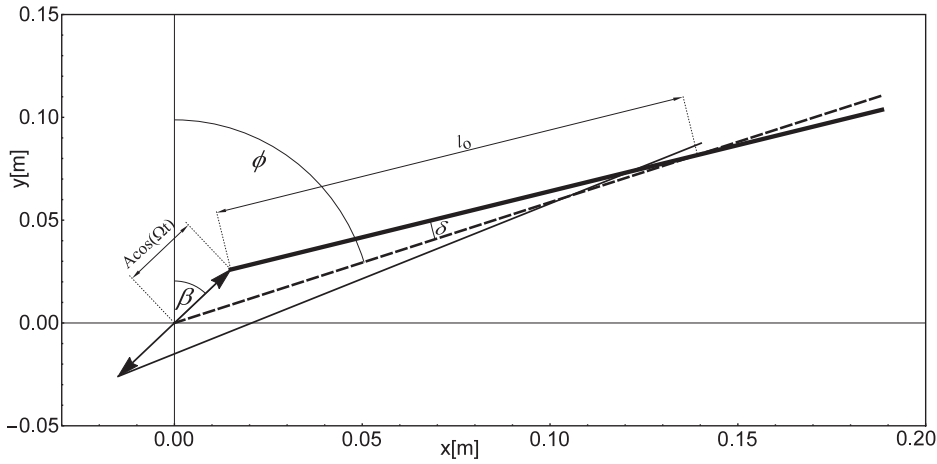


Fig. 5. Pendulum positions.

3. Approximation solution

The Eq. (1) is too complex to be solved in an analytical way but the function of the pendulum motion $\theta(t)$ can be found using the approximation of Eq. (2):

$$\theta_{\text{apr}}(t) = \phi_{\text{apr}}(t) + \delta_{\text{apr}}(t) \quad (7)$$

3.1. Undamped pendulum

The pendulum performs small oscillations around the equilibrium position ϕ_s , therefore the approximate equation of pendulum "smooth" motion ϕ_{apr} can be determined by expanding the Eq. (3) in the first-order Taylor series in $\phi = \phi_s$ for $\beta = \beta_s$ (β_s defined by Eq. (6)). The result of this expansion is a harmonic oscillator differential equation :

$$\ddot{\phi}_{\text{apr}} = (\phi_s - \phi_{\text{apr}}) \left(\frac{3}{8l^2} A^2 \Omega^2 \sqrt{9 - \frac{64g^2 l^2 \sin^2(\phi_s)}{A^4 \Omega^4}} - \frac{3g}{2l} \cos(\phi_s) \right) \quad (8)$$

whose solution is:

$$\phi_{\text{apr}} = \phi_s + A_\phi \cos(\omega_{0\text{apr}} t) \quad (9)$$

where:

$$A_\phi = \phi_0 - \phi_s, \quad \phi_0 = \phi_{\text{apr}}(t = 0), \quad \omega_{0\text{apr}} = \sqrt{\frac{3}{8l^2} A^2 \Omega^2 \sqrt{9 - \frac{64g^2 l^2 \sin^2(\phi_s)}{A^4 \Omega^4}} - \frac{3g}{2l} \cos(\phi_s)}$$

To determine the approximation of small oscillations δ , some geometric relationships occurring during the pendulum motion can be used.

Fig. 5 presents the position of the pendulum during one cycle of oscillations of the pendulum suspension point. The dashed line represents the average position of the pendulum described by ϕ . The bold line describes the position of the pendulum when the suspension point is up, and the normal line when the suspension point is down. Knowing that the δ angle is small, it can be determined using the law of sines:

$$\delta = \frac{A}{l_0} \cos(\Omega t) \sin(\phi - \beta). \quad (10)$$

The value of l_0 may be approximate as Butikov did [27] for Kapitza's mathematical pendulum, i.e. $l_0 \approx l$ (l - pendulum length) however, such an approximation may not be accurate because l_0 actually depends on all the system parameters. According to the Eq. (10) it can be said that the accuracy of determination of l_0 affects the accuracy of amplitude of small oscillations and does not affect the frequencies of these vibrations and thus the frequency spectrum. Assuming that $\phi \approx \phi_{\text{apr}}$ which is described by the Eq. (9) and using the trigonometric identity, the approximation of the δ can be expressed as:

$$\delta_{\text{apr}} = \frac{A}{l_0} \cos(\Omega t) \left(\cos(\phi_{\beta 0}) \sin(A_\phi \cos(\omega_{0\text{apr}} t)) + \sin(\phi_{\beta 0}) \cos(A_\phi \cos(\omega_{0\text{apr}} t)) \right) \quad (11)$$

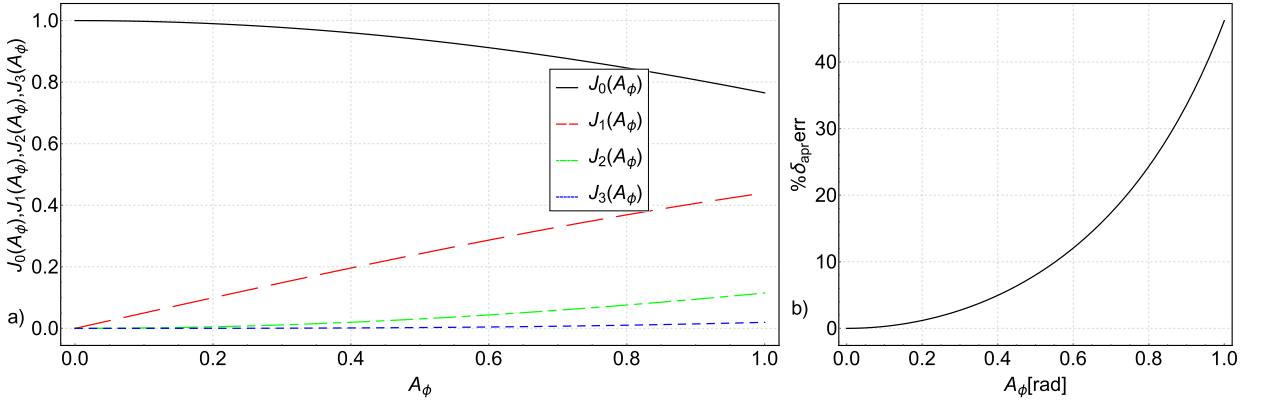


Fig. 6. a) Bessel functions of the first kind, b) Percentage error of δ_{apr} .

where $\phi_{\beta 0} = \phi_s - \beta$. The expressions $\sin(A_\phi \cos(\omega_{0apr} t))$ and $\cos(A_\phi \cos(\omega_{0apr} t))$ may be expressed by the JacobiAnger expansions [28]:

$$\begin{aligned} \sin(A_\phi \cos(\omega_{0apr} t)) &= -2 \sum_{n=1}^{\infty} (-1)^n J_{2n-1}(A_\phi) \cos((2n-1)\omega_{0apr} t) \\ \cos(A_\phi \cos(\omega_{0apr} t)) &= J_0(A_\phi) + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(A_\phi) \cos(2n\omega_{0apr} t) \end{aligned} \quad (12)$$

where $J_n(A_\phi)$ is the Bessel function of the first kind. The values of the Bessel functions of the first kind strongly depend on its order n , as shown in Fig. 6(a).

The argument of the Bessel functions in the considered case is the amplitude A_ϕ of the "smooth" oscillations of the pendulum. Due to the fact that its value is much less than 1.0 (in the example shown in Fig. 2 it is about 0.17 rad), the JacobiAnger expansions can be limited to terms to contain only J_0 and J_1 . Taking into account such an approximation, δ_{apr} takes the form:

$$\delta_{apr} = \frac{A}{l_0} J_0(A_\phi) \sin(\phi_{\beta 0}) \cos(\Omega t) + \frac{A}{l_0} J_1(A_\phi) \cos(\phi_{\beta 0}) \left(\cos(\omega_{\Omega} t) + \cos(\omega_{\Omega} t) \right) \quad (13)$$

where $\omega_{\Omega} = \Omega - \omega_{0apr}$ and $\omega_{\Omega} = \Omega + \omega_{0apr}$.

To estimate the accuracy of δ_{apr} approximation, the percentage error of truncation of the Jacobi-Anger series was determined. According to the Eq. (12), the JacobiAnger expansion is the sum of cosines whose amplitudes depend on the Bessel functions, therefore the percentage error of δ_{apr} can be approximate as a sum of percentage errors of Bessel functions truncation:

$$\% \delta_{apr} err = \frac{| -2 \sum_{n=2}^{\infty} (-1)^n J_{2n-1}(A_\phi) |}{| -2 \sum_{n=1}^{\infty} (-1)^n J_{2n-1}(A_\phi) |} + | 1 - J_0(A_\phi) / \cos(A_\phi) | \quad (14)$$

where, according to Bessel summation formula: $| 1 - J_0(A_\phi) / \cos(A_\phi) | = \frac{| 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(A_\phi) |}{| J_0(A_\phi) + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(A_\phi) |}$. Unfortunately, the expressions in the Eq. (14) dependent on $J_{2n-1}(A_\phi)$ cannot be shortened, but for $n = 20$ and $A_\phi < 1$ it can be said that the numerical value of the sums of this expressions is almost identical to the value for $n = \infty$. The relationship between the percentage error $\% \delta_{apr} err(n = 20)$ and the amplitude A_ϕ is shown in Fig. 6(b). As can be seen in Fig. 6(b), the percentage error $\% \delta_{apr} err(n = 20)$ strongly depends on the amplitude A_ϕ , but what is important for small amplitudes it is small (for $A_\phi < 0.2$ rad is less than 2 %).

According to the Eq. (7), the approximate pendulum position θ_{apr} is described as a sum of equations (9) and (13). As the approximate equation indicates, the pendulum motion during dynamic stabilization is a combination of four periodical motions with frequencies:

$$\begin{cases} f_{\omega_0} = \omega_{0apr} / 2\pi \\ f_{\Omega} = \Omega / 2\pi \\ f_{\omega_0 \Omega} = (\Omega - \omega_{0apr}) / 2\pi \\ f_{\Omega \omega_0} = (\Omega + \omega_{0apr}) / 2\pi \end{cases} \quad (15)$$

It is worth noting that in the case of the "classic" Kapitza pendulum $\phi_{\beta 0} = 0$ in the Eq. (13), so the Kapitza pendulum motion is a combination of the three oscillations: f_{ω_0} , $f_{\omega_0 \Omega}$ and $f_{\Omega \omega_0}$.

The approximate solution of θ_{apr} was checked in the example presented in Fig. 2. The value of $l_0 = 0.135$ m was determined from the intersection of the pendulum line in the maximum deflection with the line in the average position of the pendulum (see Fig. 5). The results of this check are presented in Fig. 7.

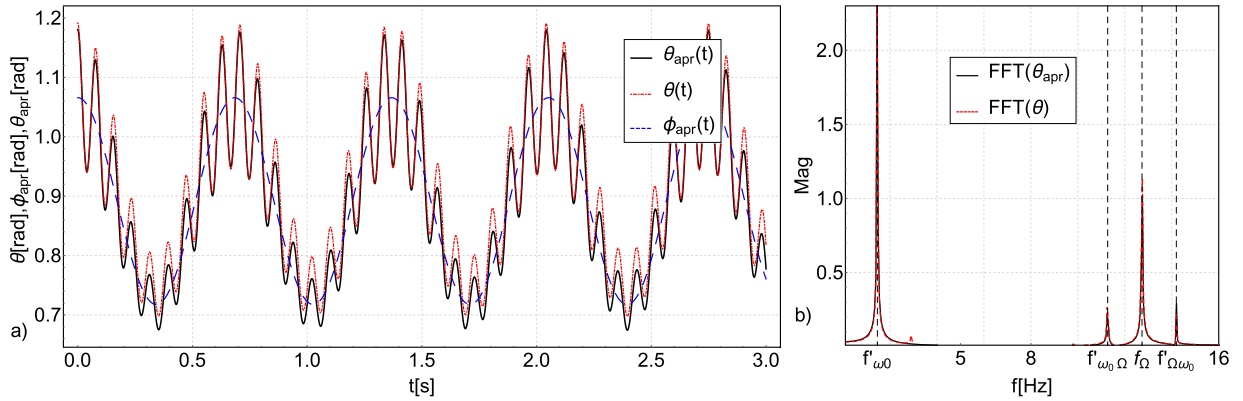


Fig. 7. The approximate vs. “real” solution of pendulum motion: a) time domain, b) frequency domain.

Table 1
Frequency spectrum comparison.

	f_{ω_0}	$f_{\omega_0\Omega}$	f_{Ω}	$f_{\Omega\omega_0}$
	[Hz]			
Analytical Approx.	1.46	11.27	12.73	14.19
FFT(Approx.)	1.45	11.25	12.75	14.20
FFT(Simulation)	1.45	11.25	12.75	14.20

The difference between the numerical solution and the analytical approximation of the pendulum position was determined using the Mean Absolute Percentage Error (MAPE). This percentage error taken from the first 10 seconds of the simulation was equal to: $MAPE \approx 2.43\%$, which indicates a good approximation of the pendulum motions. However, the frequency spectrum indicates a very good agreement between the simulation and approximation as shown in Table 1 which contains the analytical approximated frequencies (according to eq. (15)) and frequencies obtained from FFT transform. The FFT was performed in the first 20 seconds of the simulation with a sampling rate of 50 Hz which gives a frequency resolution of 0.05 Hz.

As mentioned earlier, the accuracy of the time domain results depends on the l_0 in the Eq. (10). The results presented in Table 1 indicate that l_0 does not affect the accuracy of the results in the frequency domain.

3.2. Damped pendulum

The results presented in Section 3.1 illustrate the behavior of a conservative system, i.e. without energy dissipation. In the real world, friction forces almost always cannot be eliminated (e.g. bearings in mechanisms introduce friction). The mathematical model of the pendulum, which takes into account the friction force in the form of viscous damping force takes the form:

$$\ddot{\theta} = \frac{3(A\Omega^2 \sin(\beta - \theta) \cos(t\Omega) + g \sin(\theta))}{2l} - \frac{3b}{l^2 m} \dot{\theta} \quad (16)$$

where b is the viscous damping coefficient. It should be emphasized that the damping force does not change the stability of the pendulum, just as the damping force does not change the stability of the damped harmonic oscillator. However, the parameters of the approximate solution of the pendulum motion are changed: the natural frequency of the approximated “smooth” motion of the pendulum becomes $\omega'_{0apr} = \sqrt{\omega_{0apr}^2 - (\frac{3b}{2l^2 m})^2}$ where ω_{0apr} is the natural frequency of the approximated “smooth” motion of the undamped pendulum, while the amplitude of the “smooth” approximated oscillations becomes $A'_\phi = A_\phi \exp(-\frac{3b}{2l^2 m} t)$ where A_ϕ is the amplitude of the “smooth” approximated oscillations of the undamped pendulum (see Eq. (9)). The described changes of parameters occurs analogically as in the case of the transition from an undamped oscillator to a damped one. By analogy to the undamped pendulum, the approximate motion of a damped pendulum can be described by the equations:

$$\phi_{apr} = \phi_{0apr} + A'_\phi \cos(\omega'_{0apr} t) \quad (17)$$

and

$$\delta_{apr} = \frac{A}{l_0} J_0(A'_\phi) \sin(\phi_{\beta 0}) \cos(\Omega t) + \frac{A}{l_0} J_1(A'_\phi) \cos(\phi_{\beta 0}) \left(\cos(\omega'_{\omega_0 \Omega} t) + \cos(\omega'_{\Omega \omega_0} t) \right) \quad (18)$$

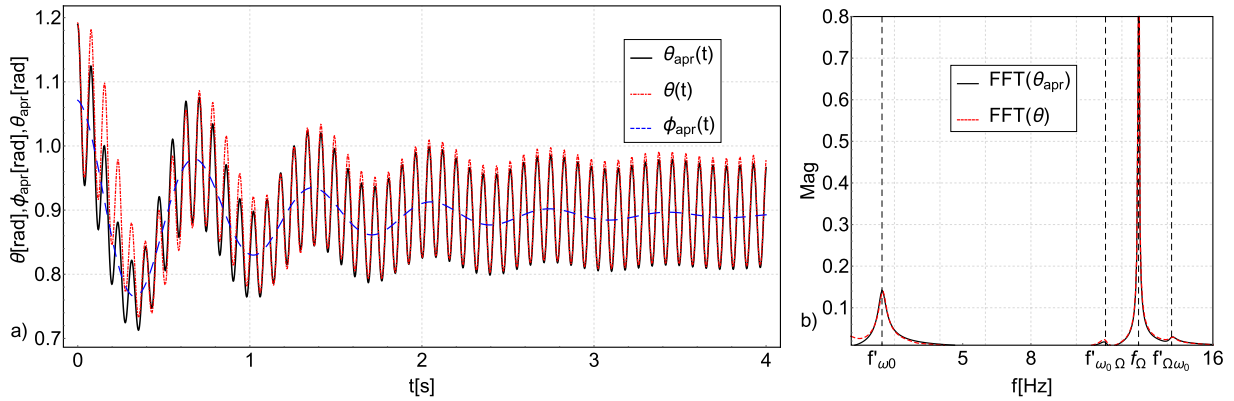


Fig. 8. The approximate vs. “real” solution of damped pendulum motion: a) time domain, b) frequency domain.

Table 2
Frequency comparison of damped pendulum.

	f'_{ω_0} [Hz]	$f'_{\omega_0\Omega}$	f_{Ω}	$f'_{\Omega\omega_0}$
Analytical Approx.	1.45	11.28	12.73	14.18
FFT(Approx.)	1.45	11.20	12.75	14.25
FFT(Simulation)	1.50	11.20	12.75	14.25

were $\omega'_{\omega_0\Omega} = \Omega - \omega'_{0apr}$ and $\omega'_{\Omega\omega_0} = \Omega + \omega'_{0apr}$. According to Eqs. (17) and (18), the damped pendulum motion is a combination of four periodical motions with frequencies:

$$\begin{cases} f'_{\omega_0} = \omega'_{0apr}/2\pi \\ f_{\Omega} = \Omega/2\pi \\ f'_{\omega_0\Omega} = (\Omega - \omega'_{0apr})/2\pi \\ f'_{\Omega\omega_0} = (\Omega + \omega'_{0apr})/2\pi. \end{cases} \quad (19)$$

Numerical verification of the approximate solution is shown in Fig. 8 and Table 2 when the pendulum with mass $m = 0.1$ kg was suppressed by the friction force with coefficient $b = 0.0025$ kg m²/s. As can be seen in Fig. 8(a), the oscillations of the pendulum decrease with time. The Mean Absolute Percentage Error from the first 10 seconds of the simulation in this case was equal to: $MAPE \approx 0.88\%$. In the frequency domain, however, definite reversal the proportion of the magnitude frequencies f_{ω_0} and f_{Ω} compared to undamped pendulum can be seen. In addition, the magnitude of frequencies $f'_{\omega_0\Omega}$ and $f'_{\Omega\omega_0}$ have significantly decreased. This phenomenon can be easily explained by analysing Eqs. (17) and (18). The amplitude A'_{ϕ} in Eq. (17) decreases exponentially with time, so the magnitude of f'_{ω_0} (natural “smooth” motion) also decreases. The Bessel function $J_1(A'_{\phi})$ in Eq. (18) decreases with time while the function $J_0(A'_{\phi})$ approaches to 1. Such changes cause that all, but one pendulum’s oscillation components are suppressed and ultimately the pendulum moves only with a one frequency f_{Ω} .

4. Experimental setup

For the purposes of the experiments which verify the process of dynamic stabilization of the pendulum in any desired position, the pendulum system which allows to change the direction of oscillations of the pendulum suspension point was projected and constructed. The designed system allows to measure the pendulum position θ as well as to measure and control the direction of oscillations of the pendulum suspension point (angle β). The mechanism of oscillations of the pendulum suspension point was driven by a DC motor DCX26L EB SL 24V [29] with planetary gearhead GPX26 A 3.9:1 and encoder ENC 30 HEDS 5540 500IMP (see ⑥ in Fig. 9). To convert the rotational motion of the motor into linear motion ($A\cos(\Omega t)$) of the suspension point the Scotch Yoke was used (see ② in Fig. 9). The Scotch Yoke (also known as the slotted link mechanism) contrary to the connecting rod, allows to get linear sinusoidal motion what is very important in this case. The direction of oscillations of the pendulum suspension point is controlled by a second DC motor DCX26L GB SL 24V [30] with planetary gearhead GPX32 A 150:1 and encoder ENC 30 HEDS 5540 500IMP (see ⑤ in Fig. 9).

To measure the pendulum position the small non-contact rotary encoder RM08 ④ with SPI interface was used [31]. The mechanism of oscillations of the pendulum suspension point together with the DCX26L EB motor and Scotch Yoke were mounted on the steel plate ③. The rotation of the steel plate by the DCX26L GB motor changes the direction of oscillations

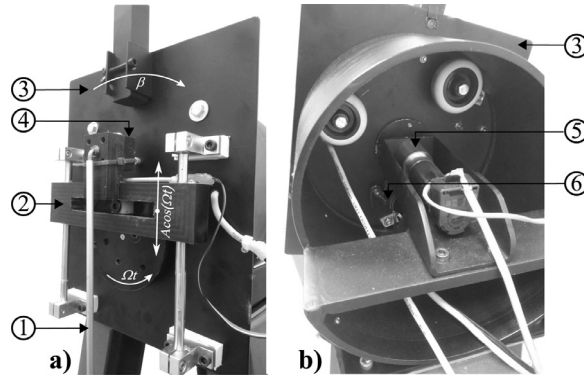


Fig. 9. Pendulum mechanism: a) front view, b) back view.

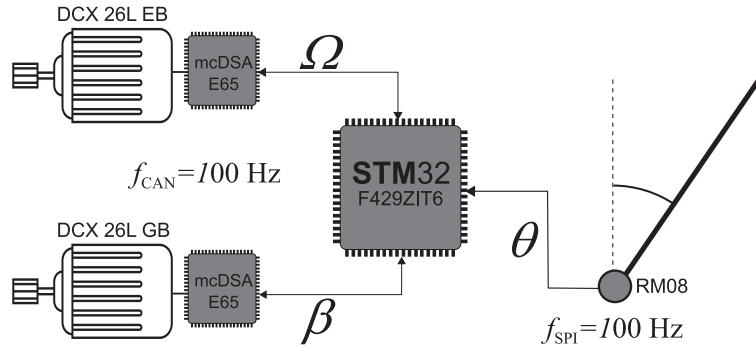


Fig. 10. Hardware schema.

Table 3
Pendulum parameters.

Parameter name	Symbol	Value	Unit
Pendulum mass	m	0.1	kg
Pendulum length	l	0.19	m
Oscillation amplitude of the suspension point	A	0.03	m
Angular frequency of the suspension point	Ω	80	1/s
Viscous damping coefficient	b	0.00037	kg m ² /s

of the pendulum suspension point β . The pendulum suspended on the Scotch Yoke was made of a homogeneous aluminum rod ①. To assemble the pendulum the ball bearings were used.

The most important parameters of the pendulum mechanism used in the experiment are listed in Table 3.

The DC motors were managed by two mcDSA-E65 [32] controllers. The operation of the motor drivers was supervised by a microcontroller STM32F429ZIT6 [33] via the CAN bus. The frequency of the motor control loop was 100 Hz and the frequency of recording the measurement data 50 Hz. The hardware schema of the system can be seen in Fig. 10. On the measuring stand described above a series of experiments were carried out, which can be grouped into the following scenarios:

4.1. Experiment I scenario

The first experiment consisted of examining the pendulum's motion when the direction of oscillations of the suspension point was fixed. Two cases were studied for which the direction of oscillations β was equal to:

- Experiment I.1: $\beta = \pi/6$ rad
- Experiment I.2: $\beta = \pi/4$ rad.

To observe the oscillations of the pendulum around its equilibrium position, the pendulum was gently manually disturbed during experiment.

4.2. Experiment II scenario

In the second experiment the dynamic stabilization of the pendulum in a moving potential well was tested.

As can be seen in Fig. 3 the point of minimum effective potential of the system (and thus also its equilibrium position) strongly depends on the direction of oscillations of the pendulum suspension point described by β . It can therefore be concluded that due to the change of the β the pendulum will be "dragged" in the potential well. Mathematical model for such a control system takes the form:

$$\begin{cases} \ddot{\theta} = \frac{3(A\Omega^2 \sin(\beta - \theta) \cos(t\Omega) + g \sin(\theta))}{2l} \\ \dot{\beta} = u \end{cases} \quad (20)$$

where u is a control input. The detailed analysis of the dynamic stabilization of the pendulum in a moving potential well can be found in [24] and also in Section Appendix B.

During this experiment the direction of oscillations of the pendulum suspension point was changed over time with constant angular velocity u . The initial value of the β was equal to 0, the final value was $\pi/4$. Again, two cases were examined for which the control input u was equal to:

- Experiment II.1: $u=0.1$ rad/s
- Experiment II.2: $u=0.6$ rad/s.

4.3. Experiment III scenario

In this experiment the possibility of active damping of the pendulum during dynamic stabilization was examined. In order to suppress the pendulum which is thrown out of its equilibrium position the full state feedback control was used. In this case, the process control consists in changing the angle of oscillations of the pendulum suspension point depending on the current system state.

Due to the fact that the state-space representation of the system described by Eq. (20) is a nonlinear and worse, depends on time explicitly, to analytically determination the control law the equation describing "smooth" motion has been used. In this case the control system is as follow:

$$\begin{cases} \ddot{\phi} = \frac{9A^2\Omega^2 \sin(2(\beta - \phi))}{16l^2} + \frac{3g \sin(\phi)}{2l} \\ \dot{\beta} = u. \end{cases} \quad (21)$$

In the next step, the Eq. (21) was linearised around a desired operating point in $\phi = \phi_s$ and $\beta = \beta_s$. For such a linearised system, a linear-quadratic regulator (LQR) could already be used to determine the full state feedback control. The experiment considered the case when the pendulum suspension point oscillates in the vertical direction, so the desired operating point was $\phi_s = 0$ and $\beta_s = 0$. To meet the requirements of the LQR regulator for "smooth" motion, the input signal u was dependent on the average state variables, i.e. $\bar{\theta}$ and $\bar{\dot{\theta}}$, and on the current β position. The averaged pendulum position $\bar{\theta}$ was determined using a moving average for the period of fast oscillations of the pendulum suspension point. The period of fast oscillation of the pendulum suspension point was equal $T = 2\pi/\Omega \approx 0.08$ s while the measurement period $\Delta t = 0.02$ s, therefore four consecutive measurement samples were taken for the averaging. The average pendulum velocity $\bar{\dot{\theta}}$ was calculated using first-order divided difference of $\bar{\theta}$. The control equation in Eq. (21) can therefore be written as a feedback state equation:

$$\dot{\beta} = k_1 \bar{\theta} + k_2 \bar{\dot{\theta}} + k_3 \beta. \quad (22)$$

The regulator gains k_1 , k_2 , k_3 have been manually tuned to values: $k_1 = 2.91$ 1/s, $k_2 = 0.2$, $k_3 = 15.75$ 1/s. More details about damping of the pendulum during dynamic stabilization can be found in [25] and also in Section Appendix C.

4.4. Experiment IV scenario

Finally, the transition of the pendulum through the whole range of its angle position was tested. The angular velocity of β during the transition was equal to $\dot{\beta} = 0.2$ rad/s.

5. Results

This section presents the results obtained during the experiments I, II, III and IV. The first step to analyse the experimental results was to determine the viscous damping coefficient b presented in Table 3. This coefficient was determined on the basis of the measurement data collected in experiment I.1. The data was "smoothed" due to fast oscillations by moving average filter and then fitted to the theoretical curve described by the Eq. (17) (see Fig. 11). The value of the determined coefficient was $b = 0.00037$ kg m²/s.

Table 4
Experiment I.1 results.

	f'_{ω_0}	$f'_{\omega_0 \Omega}$	f_{Ω}	$f'_{\Omega \omega_0}$	ϕ_s
	[Hz]				[°]
Approximation	1.46	11.27	12.73	14.19	51.1
Experiment	1.50	11.10	12.60	14.10	51.0
Simulation	1.50	11.20	12.70	14.30	51.4

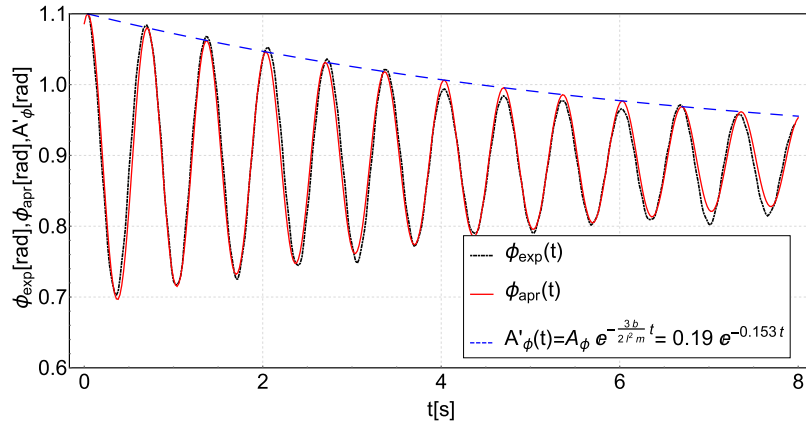


Fig. 11. Determination of the coefficient b .

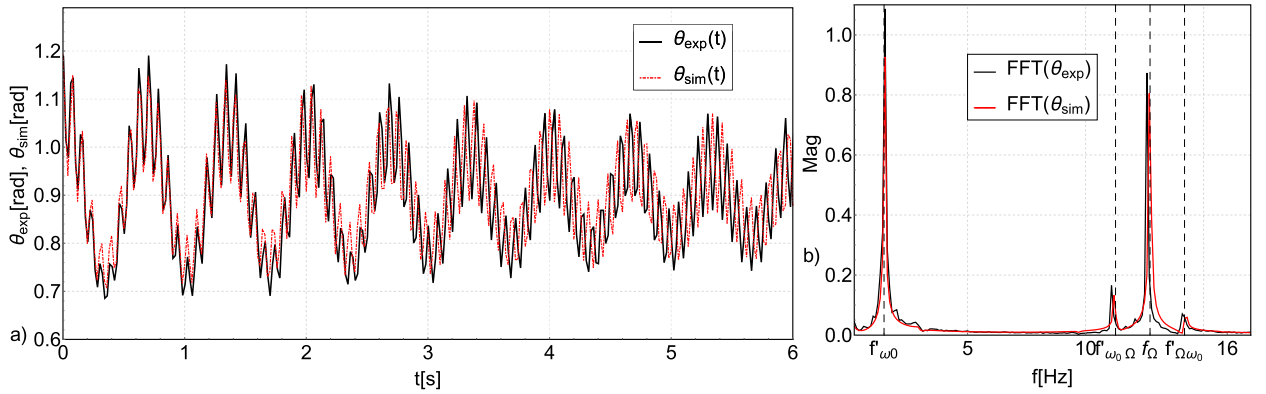


Fig. 12. Experiment vs. simulation - experiment I.1: a) time domain, b) frequency domain.

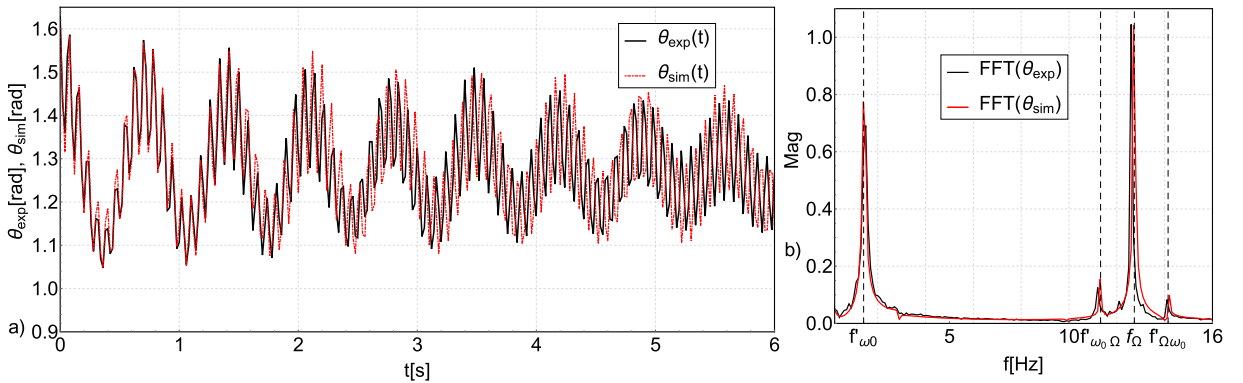


Fig. 13. Experiment vs. simulation - experiment I.2: a) time domain, b) frequency domain.

Table 5

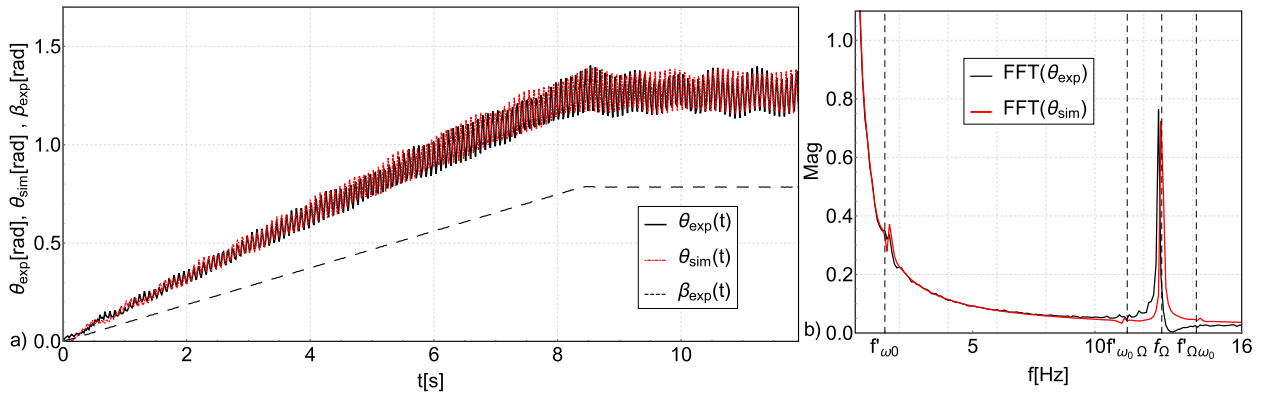
Experiment I.2 results.

	f'_{ω_0}	$f'_{\omega_0\Omega}$	f_{Ω}	$f'_{\Omega\omega_0}$	ϕ_s
	[Hz]				[°]
Approximation	1.41	11.32	12.73	14.14	72.7
Experiment	1.50	11.20	12.60	14.10	72.9
Simulation	1.40	11.30	12.70	14.20	73.2

Table 6

Experiment II.1 results.

	f'_{ω_0}	$f'_{\omega_0\Omega}$	f_{Ω}	$f'_{\Omega\omega_0}$	ϕ_s
	[Hz]				[°]
Approximation	1.41	11.32	12.73	14.14	72.7
Experiment	1.40	-	12.60	-	72.1
Simulation	1.40	-	12.70	-	72.7

**Fig. 14.** Experiment vs. simulation - experiment II.1: a) time domain, b) frequency domain.

5.1. Experiment I results

The video material recorded during the experiment I is available at https://www.youtube.com/watch?v=MPduMcy_q0Q. The measurement data from the experiments I.1 and I.2 and its comparison with simulation results are presented in Fig. 12, Table 4 and Fig. 13, Table 5 respectively. The approximated ϕ_s presented in Tables 4 and 5 were calculated from the first equation in (6), while the simulated and experimental ϕ_s were calculated as the average values of the pendulum position (the pendulum oscillates around its equilibrium position ϕ_s). The FFT transform was performed in the first 10 seconds of the simulation/experiment with a sampling rate of 50 Hz (the same as the measurement recording rate) which gives a frequency resolution of 0.1 Hz.

Based on the results of experiment I, the following conclusions can be drawn:

- the process of dynamic stabilization of the pendulum in a desired position in the case when the pendulum suspension point oscillates in a straight line has been fully confirmed in the experiment.
- the experimental results are in line with the numerical simulation.
- the quantitative parameters describing the phenomenon of dynamic stabilization obtained by the analytical approximation are consistent with the experimental results.

Emerging small differences between experimental and numerical data can be caused by various factors e.g. measurement uncertainty of the system parameters, inaccuracy in control of the angular velocity Ω or the occurrence of some physical phenomena which were omitted in the mathematical model of the pendulum system.

5.2. Experiment II results

In this experiment the pendulum motion was studied when the direction of oscillations of the pendulum suspension point varied linearly in time to a predetermined value. The video material from the experiment II is available at https://www.youtube.com/watch?v=Df8Gp_vqtpM. The results of this experiment are shown in Fig. 14, Table 6 and Fig. 15, Table 7 respectively. The value of ϕ_s was calculated from the moment when $\beta = \pi/4$. The lack of explicit components of

Table 7
Experiment II.2 results.

	f'_{ω_0}	$f'_{\omega_0\Omega}$	f_{Ω}	$f'_{\Omega\omega_0}$	ϕ_s
	[Hz]				[°]
Approximation	1.41	11.32	12.73	14.14	72.7
Experiment	1.60	-	12.60	-	72.8
Simulation	1.40	-	12.70	-	72.3

Table 8
Experiment III results.

	f'_{ω_0}	$f'_{\omega_0\Omega}$	f_{Ω}	$f'_{\Omega\omega_0}$	ϕ_s
	[Hz]				[°]
Approximation	1.61	11.12	12.73	14.34	0
Experiment	1.60	11.00	-	14.2	0.3
Simulation	-	-	-	-	0

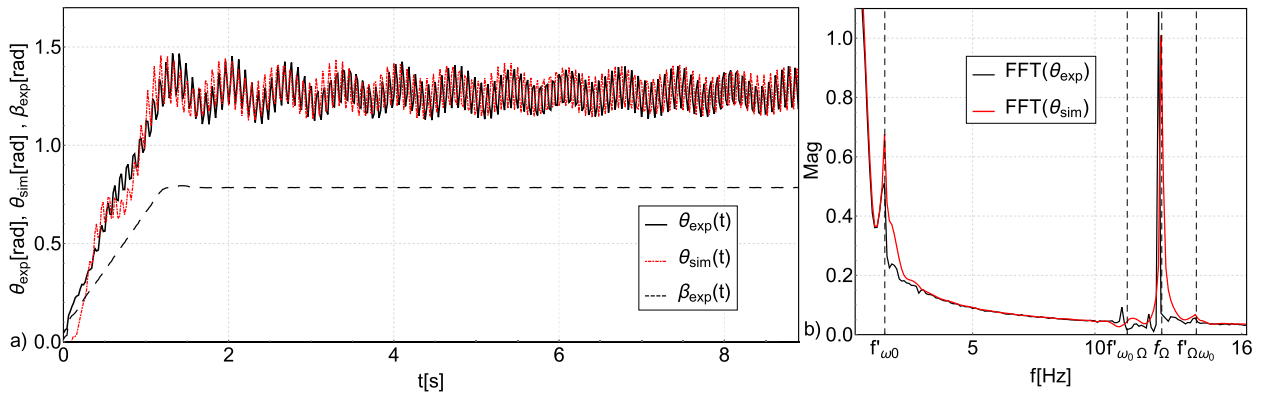


Fig. 15. Experiment vs. simulation - experiment II.2: a) time domain, b) frequency domain.

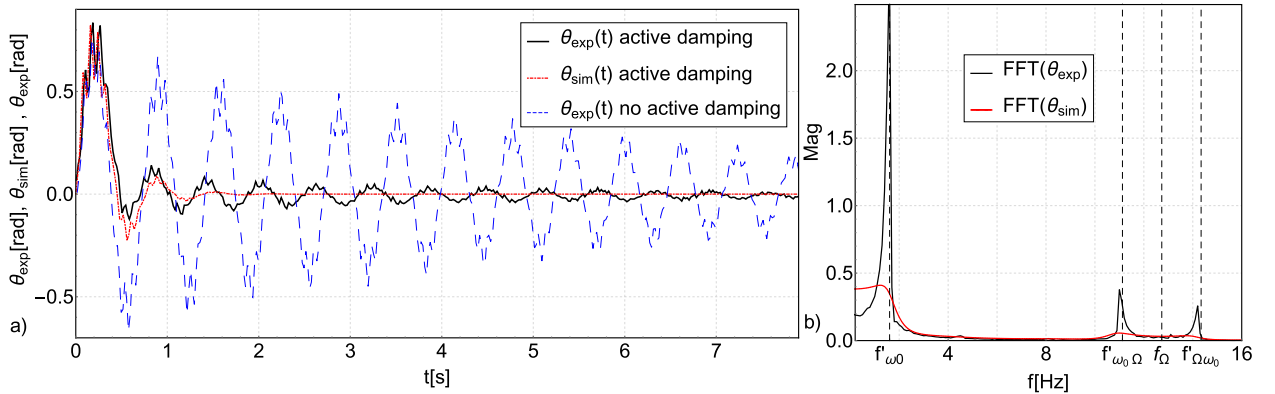


Fig. 16. Experiment vs. simulation - experiment III: a) time domain, b) frequency domain.

the $f'_{\omega_0\Omega}$ and $f'_{\Omega\omega_0}$ in the frequency spectrum of the experiment II.1 and II.2 is due to the fact that the pendulum was not actually excited, so $A'_\phi \approx 0$.

The results of the experiment II can be summarized as follows:

- the process of dynamic stabilization of the pendulum in a moving potential well has been fully confirmed in the experiment.
- experimental and numerical results are consistent with each other.
- the analytically approximated parameters of the process are consistent with the experimental results.

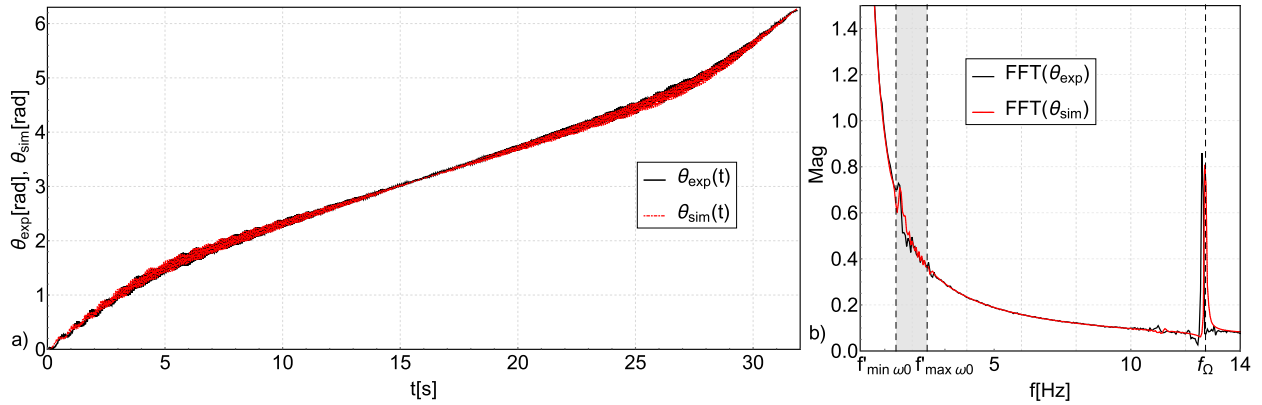


Fig. 17. Experiment vs. simulation - experiment IV: a) time domain, b) frequency domain.

5.3. Experiment III results

The video material from the experiment III where the active damping of the pendulum was studied is available at <https://www.youtube.com/watch?v=y8sGXzsWUI0>. The Fig. 16 and Table 8 presents the results of this experiment.

To compare the quality of active damping control, Fig. 16 also shows the experiment without control. It is clearly seen that active damping control reduced the oscillations of the pendulum in 0.5 seconds by about 85%. Further damping process in the experiment does not go as fast as during simulation, where the oscillations completely disappeared after about 1.5 seconds. The most likely reason for this difference were inaccuracies in the control process, specifically in precise control of angular velocity $\dot{\beta}$ which were caused by the backlash of the motor gearbox and the friction forces. Despite these errors, however, it should be recognized that the process of active damping control was successful. The lack of the FFT frequencies from the simulation (Table 8) is due to the fact that the pendulum was immediately suppressed. The absence of f_{Ω} frequency in the experimental spectrum can be explained by the approximation solution described by the Eq. (18): for $\beta \approx 0$, the parameter $\phi_{\beta 0} \approx 0$, so $\sin(\phi_{\beta 0})\cos(\Omega t) \approx 0$.

5.4. Experiment IV results

Finally, the fourth experiment shows the transition of the pendulum through the whole range of its angle position. The results of the experiment are presented in the Fig. 17 and in the video material: <https://www.youtube.com/watch?v=FWCSBlqH7II>.

As can be seen in Fig. 17, the transition of the pendulum through the whole 360° range of its angle position was confirmed by the experiment. On the spectrum plot, the area of permissible f'_{ω_0} values was marked (according to Eq. (19)). The range of these frequencies according to the analytical approximation, is: $f'_{\omega_0} \in \langle 1.41, 2.55 \rangle \text{ Hz}$. Slight disturbances in this area, visible in both experiment and simulation, are due to the fact that the pendulum during the transition is gently excited from its equilibrium state. The experiment IV also indirectly proves the possibility of stabilizing the pendulum in any position.

6. Conclusion

In this paper the experimental results of dynamic stabilization of the pendulum in the case when the pendulum suspension point oscillates in a straight line have been presented. These results proved to be consistent with the previous numerical and analytical studies. An important factor in the success of the experiments was the test stand, which allowed to measure and control the pendulum stabilization process.

The work also contains an analytical approximation of the pendulum motion during dynamic stabilization. The quantitative results i.e. the frequency spectrum obtained from approximate analytical solution, were found to be largely consistent with the experimental data.

Summarizing the results of this work, it can be said that when the pendulum suspension point oscillates in a straight line the following facts can be stated:

- an approximate solution of the pendulum motion during dynamic stabilization can be obtained using the JacobiAnger expansion.
- it is possible to stabilize the pendulum in any position.
- by changing the direction of oscillations of the pendulum suspension point, it is possible to transfer the pendulum from point to point.
- the excessive pendulum oscillations can be suppressed using the active damping control.

An interesting issue is the transition of the pendulum from point to point. This process was experimentally and numerically tested (see experiment *II*) but only for cases where the direction of oscillations of the pendulum suspension point changed with a constant angular velocity, which took only two values $\dot{\beta} = 0.1$ rad/s and $\dot{\beta} = 0.6$ rad/s. One can ask the question how to carry out such a process in the shortest time, what is the time-optimal control? It can be assumed that the time-optimal control will require a very precise control of $\dot{\beta}$, which as can be seen in the experiment *III*, has not been sufficiently achieved. Therefore, the further studies on dynamic stabilization of the pendulum will focus on improving the quality of $\dot{\beta}$ control and, in the longer term, finding the time-optimal control for the pendulum transition problem.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Maciej Cieřkowski: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Visualization, Investigation.

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Appendix A. Determining the equation of "smooth" pendulum motion

Substituting Eq. (2) into (1) and expanding the result in the first-order Taylor series with respect to small oscillations $\delta(t)$, the equation (1) can be rewritten in approximation form:

$$\ddot{\phi} + \ddot{\delta} = -\frac{3A\delta\Omega^2 \cos(\beta - \phi) \cos(\Omega t)}{2l} + \frac{3A\Omega^2 \sin(\beta - \phi) \cos(\Omega t)}{2l} + \frac{3g\delta \cos(\phi)}{2l} + \frac{3g \sin(\phi)}{2l}. \quad (\text{A.1})$$

Small, fast-changing oscillations δ are the result of the fast-changing force acting on pendulum suspension point causing its oscillations. The position of the pendulum suspension point depends on time according to the rule $A \cos(\Omega t)$ (see Fig. 1), therefore the acceleration of the suspension point as well as $\ddot{\delta}$ are proportional to Ω^2 . The second term of the Eq. (A.1) satisfies these condition (the first term is also proportional to Ω^2 but it depends on δ , therefore is too small to describe the acceleration $\ddot{\delta}$). Based on the above considerations, the acceleration $\ddot{\delta}$ can be presented as follows:

$$\ddot{\delta} = \frac{3A\Omega^2 \sin(\beta - \phi) \cos(\Omega t)}{2l}. \quad (\text{A.2})$$

Considering the fact that ϕ changes very slowly during the one period of fast oscillations, the δ can be obtain by time integration of the expression (A.2):

$$\delta = -\frac{3A \cos(\Omega t) \sin(\beta - \phi)}{2l}. \quad (\text{A.3})$$

Substituting the Eq. (A.3) into Eq. (A.1) and averaging the result due to the rapid oscillations ($\overline{\cos(\Omega t)} = 0$, $\overline{\cos(\Omega t)^2} = \frac{1}{2}$), the equation of "smooth" pendulum motion described by Eq. (3) is obtained.

Appendix B. Pendulum in a moving potential well

Let us consider the process of dynamic stabilization of the pendulum at point $\phi_s = 1.27$ rad, when the initial conditions are as follow: $\theta(0) = 0$ rad and $\dot{\theta}(0) = 0$ rad/s. According to Eg. (6) the direction of the vibrations of the pendulums suspension point should be equal to $\beta = \beta_s = \pi/4$ rad (exactly as final value of β in experiment *II*). The simulation results for such a conditions are presented in Fig. B.18.

As can be seen in Fig. B.18 a), the stabilization process failed - the pendulum rotated through a full 360 degrees. To understand this results, the effective potential of the system and total mechanical energy of the pendulum should be considered. The total mechanical energy of the pendulum in the case of "smooth" motion, can be presents as a sum of effective potential energy and kinetic energy of the pendulum:

$$E_{\text{mech}} = U_{\text{ef}} + \frac{1}{6} l^2 m \dot{\phi}^2. \quad (\text{B.1})$$

According to Eg. (B.1) the total mechanical energy of the pendulum at the begining of the process presented in Fig. B.18 a) is about 0.093 J ($\dot{\phi} = 0$, so only the effective potential energy is non-zero) while the maximum permissible mechanical energy of the pendulum (hereinafter called E_{critical}) trapped in the potential well presented in Fig. B.18 b) is about 0.003 J. Based on this analysis, it can be easily seen that the stabilization process failed because the mechanical energy of the

pendulum at the beginning of the process was too high to trap the pendulum in the potential well. The simplest process of dynamic stabilization of the pendulum for $\phi_s = 1.27$ rad, $\theta(0) = 0$ rad and $\dot{\theta}(0) = 0$ rad/s should therefore start from $\beta = 0$ rad (stability condition for $\phi_s = 0$) and β should be increased with time to $\beta = \beta_s = \pi/4$ rad. In general, this process can be described by the Eq. (20).

Let us consider the case when the direction of oscillations of the pendulums suspension point changes uniformly from $\beta = 0$ rad to $\beta = \beta_s = \pi/4$ rad (as in experiment II). The control input in the Eq. (20) can therefore be written as:

$$\dot{\beta} = \nu H(\beta_s - \beta) \quad (\text{B.2})$$

where H is Heaviside step function and ν is a constant. The simulation results in the form of mechanical energy of the pendulum at the end of transition for different values of constant ν are presented in Fig. B.19.

Fig. B.19 a) presents the mechanical energy of the pendulum for selected ν at the end of transition and also the extreme positions of the pendulum in its movement from one edge of a potential well to the second edge. For example in the case when $\nu = 0.95$ rad/s, the pendulum oscillates between 1.1 rad and 1.47 rad. It is clearly seen that the mechanical energy of the pendulum at the end of transition and therefore the amplitude of its oscillations around equilibrium state $\phi_s = 1.27$ rad depends on value of the control input ν . It is also clearly seen that for $\nu = 1.55$ rad/s the pendulum is no longer trapped in the potential well. Based on the simulation results it can be concluded that there is a certain critical value of ν (hereinafter called $\nu_{critical}$) above which the pendulum is no longer trapped in the potential well because its mechanical energy is greater than $E_{critical}$. Fig. B.19 b) presents numerically obtained function $E_{mech}(\nu)$ at the end of transition, and it can be seen that this relationship is not a simple monotonic dependence. The critical value of ν for considered case, corresponding to energy $E_{critical}$ was numerically designated as 1.375 rad/s. In the experiment II described in Section 5.2 the values of ν were equal to 0.1 rad/s and 0.6 rad/s so there were far from the $\nu_{critical}$.

Appendix C. Determining the control law of active damping of the pendulum

To determine the control law of active damping of the pendulum, the Eq. (21) was used. To use the LQR controller, the state-space equation must be linear, so the Eq. (21) was transformed into the state-space representation and then linearized

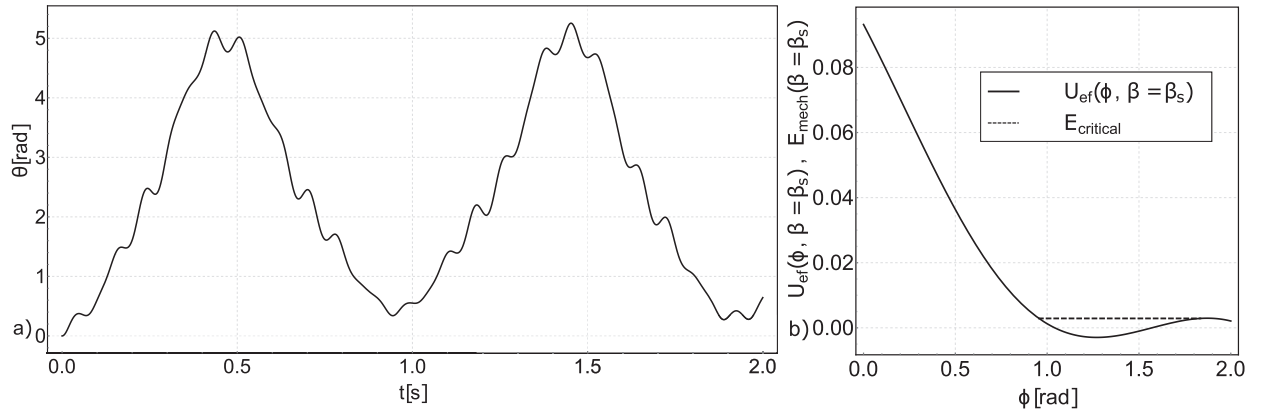


Fig. B1. Simulation results for $\beta_s = \pi/4$: a) time domain, b) critical mechanical energy of the pendulum in potential well.

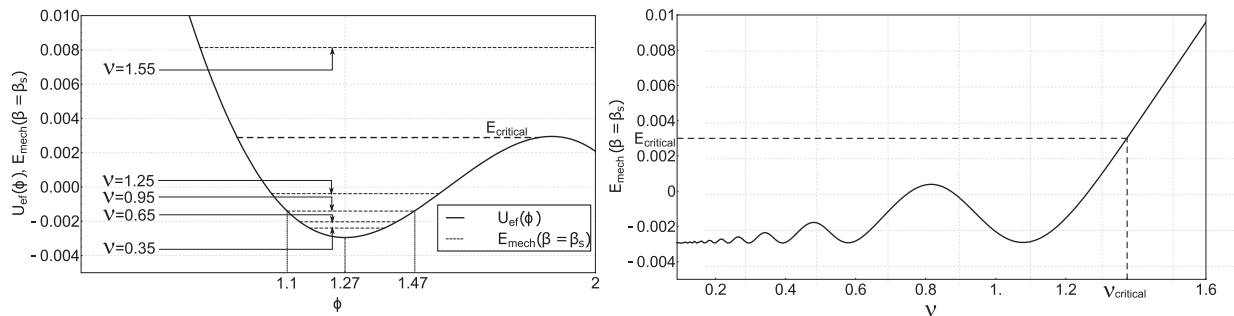


Fig. B2. Transition of the pendulum for different values of ν : a) the mechanical energy of the pendulum for selected ν , b) the mechanical energy of the pendulum as function of ν .

around the desired operating point $\phi = \phi_s$, $\dot{\phi} = 0$ and $\beta = \beta_s$:

$$\begin{pmatrix} \dot{\phi} \\ \ddot{\phi} \\ \dot{\beta} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -C_1 + C_2 & 0 & C_1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \dot{\phi} \\ \beta \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u \quad (\text{C.1})$$

where $C_1 = \frac{9A^2\Omega^2 \cos(2(\beta_s - \phi_s))}{8l^2}$, $C_2 = \frac{3g \cos(\phi_s)}{2l}$. According to the LQR theory the unknown input u is a linear function of the state vector:

$$u = k_1(\phi - \phi_s) + k_2\dot{\phi} + k_3(\beta - \beta_s) \quad (\text{C.2})$$

and the regulator gains k_1 , k_2 , k_3 are obtained using linear-quadratic regulator (see for example [34]). The "smooth" pendulum motion vector $(\phi, \dot{\phi})$ is not measured during the experiment, so to apply the Eq. (C.2) in experiment, the average measured values $(\bar{\theta}, \bar{\dot{\theta}})$ instead of $(\phi, \dot{\phi})$ have to be used as it was described in Section 4.3.

References

- [1] A. Stephenson, On a new type of dynamical stability, *Mem. Proc. Manch. Lit. Phil. Sci.* 52 (1908) 1–10.
- [2] A. Stephenson, Xx. on induced stability, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 15 (86) (1908) 233–236.
- [3] E. Lowenstern, Xxxix. the stabilizing effect of imposed oscillations of high frequency on a dynamical system, *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 13 (84) (1932) 458–486.
- [4] P.L. Kapitza, Pendulum with a vibrating suspension, *Usp. Fiz. Nauk.* 44 (1951) 7–15.
- [5] V.N. Chelomei, On the possibility of increasing the stability of elastic systems by mean of vibration, *Dokl. Akad. Nauk. SSSR* 110 (1956) 345–347.
- [6] A.A. Mailybaev, A.P. Seyranian, Stabilization of statically unstable systems by parametric excitation, *J Sound Vib* 323 (3–5) (2009) 1016–1031.
- [7] A. Seyranian, A. Seyranian, Chelomei'S problem of the stabilization of a statically unstable rod by means of a vibration, *J. Appl. Math. Mech.* 72 (6) (2008) 649–652.
- [8] A. Zelei, G. Stepan, The influence of parametric excitation on floating bodies, in: *PAMM: Proceedings in Applied Mathematics and Mechanics*, 8, Wiley Online Library, 2008, pp. 10929–10930.
- [9] F. Jafari, S. Tahmasian, S.D. Ross, J.J. Socha, Control of gliding in a flying snake-inspired n-chain model, *Bioinspiration & biomimetics* 12 (6) (2017) 066002.
- [10] C.J. Richards, T.J. Smart, P.H. Jones, D. Cubero, A microscopic kapitza pendulum, *Sci Rep* 8 (1) (2018) 13107.
- [11] T. Karino, S. Kawata, Y.J. Gu, Dynamic stabilization of plasma instability, *High Power Laser Sci. Eng.* 7 (11) (2019) e3.
- [12] I. Gilary, N. Moiseyev, S. Rahav, S. Fishman, Trapping of particles by lasers: the quantum kapitza pendulum, *J Phys A Math Gen* 36 (25) (2003) L409.
- [13] H. Saito, M. Ueda, Dynamically stabilized bright solitons in a two-dimensional bose-einstein condensate, *Phys. Rev. Lett.* 90 (4) (2003) 040403.
- [14] I.M. Arkhipova, A. Luongo, A.P. Seyranian, Vibrational stabilization of the upright statically unstable position of a double pendulum, *J Sound Vib* 331 (2) (2012) 457–469.
- [15] I. Wickramasinghe, J.M. Berg, Vibrational control of mathieu's equation, in: *2013 IEEE/ASME International Conference on Advanced Intelligent Mechatronics*, IEEE, 2013, pp. 686–691.
- [16] F. Bullo, Averaging and vibrational control of mechanical systems, *SIAM J. Control Optim.* 41 (2) (2002) 542–562.
- [17] L. Landau, E. Lifshitz, *Mechanics*, Pergamon, Oxford, 1976.
- [18] D.J. Sudor, S.R. Bishop, Inverted dynamics of a tilted parametric pendulum, *European Journal of Mechanics-A/Solids* 18 (3) (1999) 517–526.
- [19] S. Weibel, T. Kaper, J. Baillieu, Global dynamics of a rapidly forced cart and pendulum, *Nonlinear Dyn* 13 (2) (1997) 131–170.
- [20] H. Yabuno, K. Goto, N. Aoshima, Swing-up and stabilization of an underactuated manipulator without state feedback of free joint, *IEEE Transactions on Robotics and Automation* 20 (2) (2004) 359–365.
- [21] H. Yabuno, T. Matsuda, N. Aoshima, Reachable and stabilizable area of an underactuated manipulator without state feedback control, *IEEE/ASME Trans. Mechatron.* 10 (4) (2005) 397–403.
- [22] G.J. VanDalen, The driven pendulum at arbitrary drive angles, *Am J Phys* 72 (4) (2004) 484–491.
- [23] M. Cieřkowski, Stabilization of pendulum in various inclinations using open-loop control, *Acta mechanica et automatica* 5 (2011) 22–29.
- [24] M. Cieřkowski, Dynamic stabilization of the pendulum in a moving potential well., in: *MMAR*, 2016, pp. 54–58.
- [25] M. Cieřkowski, Damping of the pendulum during dynamic stabilization in arbitrary angle position, in: *Polish Control Conference*, Springer, 2017, pp. 45–54.
- [26] M.L. Gambhir, *Stability analysis and design of structures*, Springer Science & Business Media, 2013.
- [27] E.I. Butikov, On the dynamic stabilization of an inverted pendulum, *Am J Phys* 69 (7) (2001) 755–768.
- [28] M. Abramowitz, I. Stegun, *Handbook of mathematical functions*, Dover, Mineola, NY, 1965.
- [29] Maxon, Dcx 26 l precious metal brushes dc motor, 2019a, (https://www.maxongroup.com/medias/sys_master/root/8833375797278/19-EN-88.pdf).
- [30] Maxon, Dcx 26 l graphite brushes dc motor, 2019b, (https://www.maxongroup.com/medias/sys_master/root/8833375862814/19-EN-89.pdf).
- [31] RLS, Rm08 super small non-contact rotary encoder, 2020, (https://www.rls.si/en/fileuploader/download/download/?d=1&file=custom%2Fupload%2FRM08D01_14.pdf).
- [32] miControl, mcDSA-e65-modul, 2018, (https://www.micontrol.de/mc_files/datasheets/DataSheet_mcDSA-E65-Modul_en.pdf).
- [33] STMicroelectronics, Stm32f427xx stm32f429xx, 2018, (<https://www.st.com/resource/en/datasheet/dm00071990.pdf>).
- [34] A.E. Bryson, *Applied Optimal Control: Optimization, Estimation and Control*, CRC Press, 1975.